

HOW CHILDREN FEEL PAIN

*Pediatric Pain PRN
Curriculum*

The
MAYDAY
Fund

Objectives

1. Describe theories that inform our understanding of pain
2. Explain the biopsychosocial model of pain

How do you know your experience is pain?



Type your answer here.

Cartesian Theory of Pain (Descartes)

Descartes proposed that the brain directs responses to pain and pain-related behaviors.

Descartes suggested reflexive and highly predictable responses to pain are mechanistic to avoid further tissue damage.

Descartes further proposed that the intensity of pain is directly related to the severity of tissue damage.

This remains a commonly held assumption about pain.

Patients who report pain intensity disproportionate to tissue injury are often the most challenging to treat; in part, because this defies expectations and long held beliefs about how people feel pain.

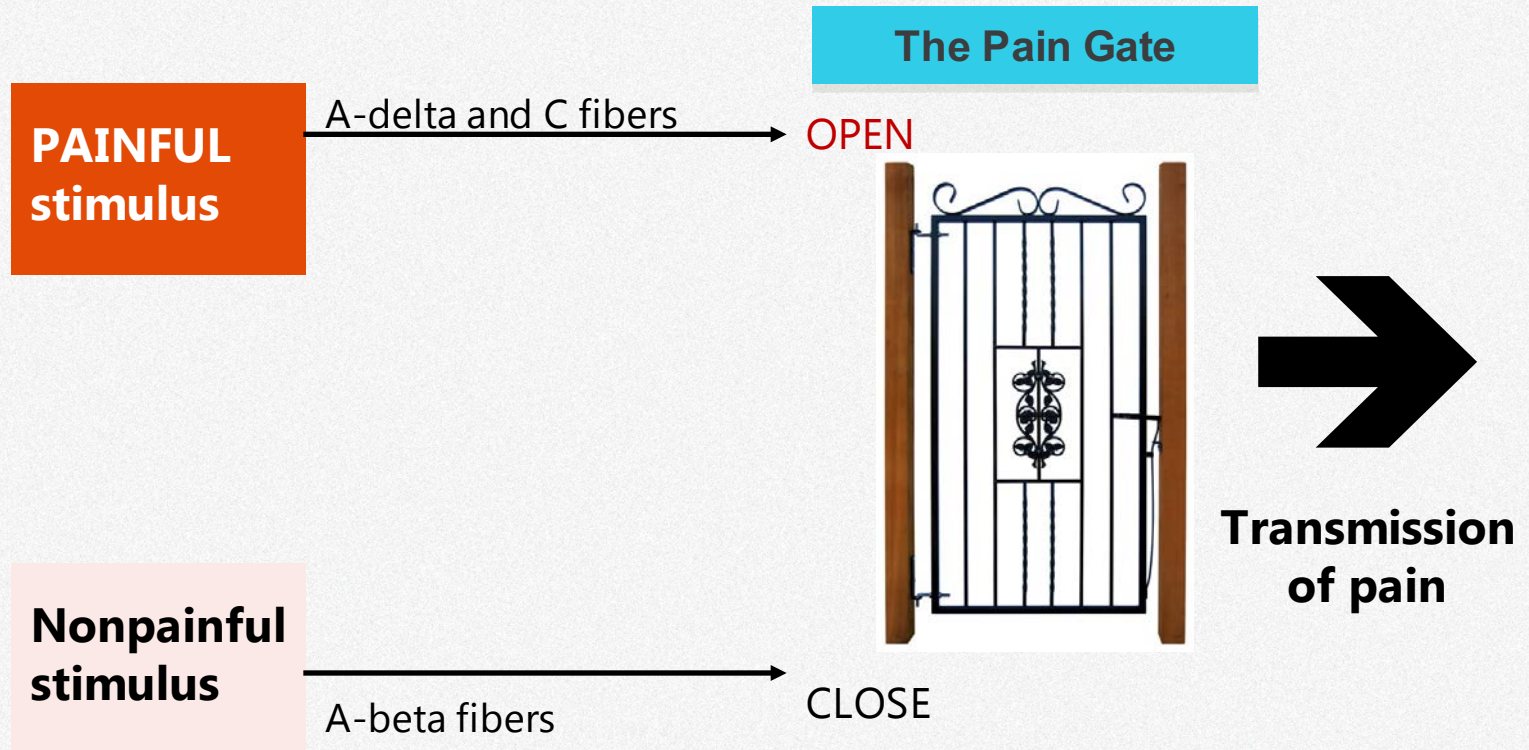
No BRAIN
No Pain



Gate Control Theory

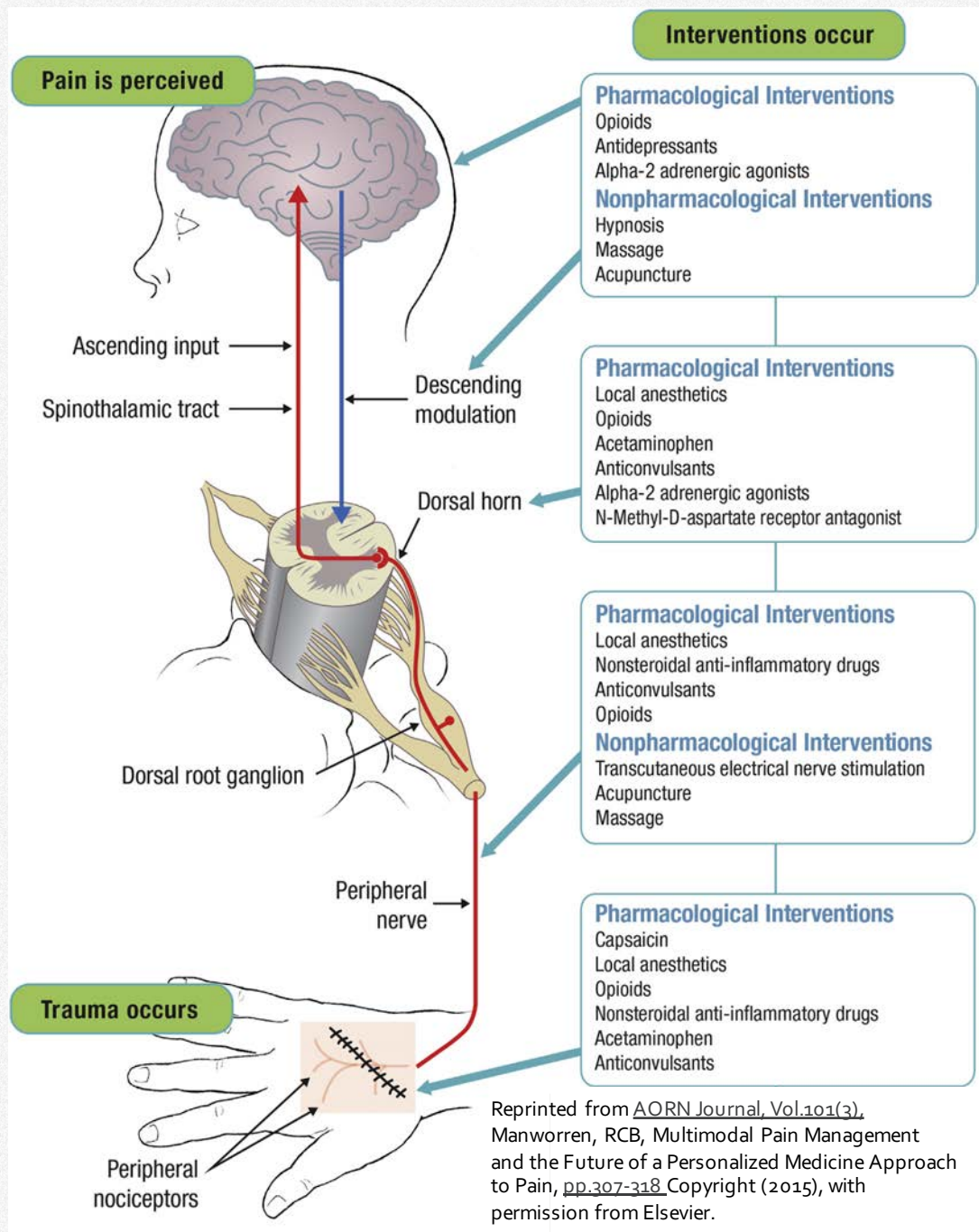
Proposes that both large (A-fibers) and small (C-fibers) synapse onto cells in the substantia gelatinosa (SG) and the 1st central transmission cells. The inhibitory effect exerted by SG cells onto the primary afferent fiber terminals at the T cells is increased by activity in A-fibers and decreased by activity in C-fibers, the opening and closing of the gate.

This remains a commonly used explanation for how a non-painful stimulus applied to a site of pain works to relieve pain. For example, rubbing a site of pain closes the gate to relieve the pain .



Multimodal Analgesia: Many Gates?

Research clearly shows that many factors influence pain perception. Pain is not simply a neurophysiological phenomenon or the opening and closing of electrochemical gates.





How do other interventions
(not listed on previous page) **work**
to relieve pain?

What is the mechanism of
action for these interventions?

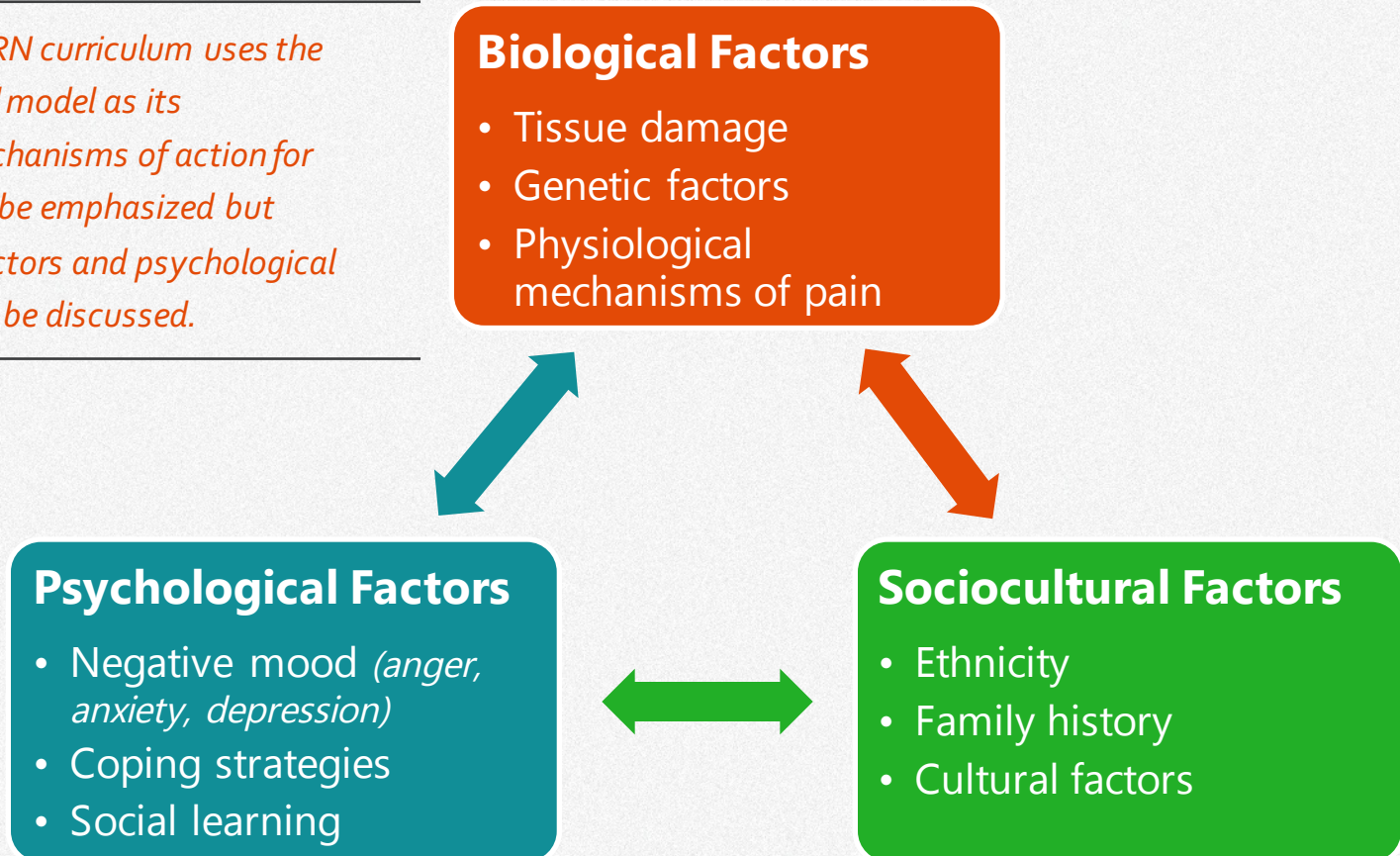
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Biopsychosocial Model of Pain

The Pediatric PRN curriculum uses the biopsychosocial model as its framework. Mechanisms of action for treatments will be emphasized but sociocultural factors and psychological factors will also be discussed.

The sensory stimulus is associated with tissue damage but the social aspects set the framework for how children react to pain. **Culture sets boundaries for how children experience, respond to, and describe pain.**

Psychological factors like personality, mood, and learned behaviors, such as coping strategies, can also influence how children experience pain and its impact on quality of life.



Appendix

References

Gate control theory

Moayedi, M. Davis, KD. (2013). *Journal of Neurophysiology*. 109 (1), 5-12.

Multimodal analgesia

Manworren, RCB, (2015). Multimodal Pain Management and the Future of a Personalized Medicine Approach to Pain, *AORN Journal*, Vol.101(3), 307-318 .

Biopsychosocial Model of Pain

Vetter T, McGwin G, Bridgewater CL, Madan-Swain A, Ascherman LI. (2013). Validation and clinical applications of a biopsychosocial model of pain intensity and functional disability in patients with pediatric chronic pain condition referred to a subspecialty clinic. *Pain Res Treat*. Doi: 10.1155/2013/143292.